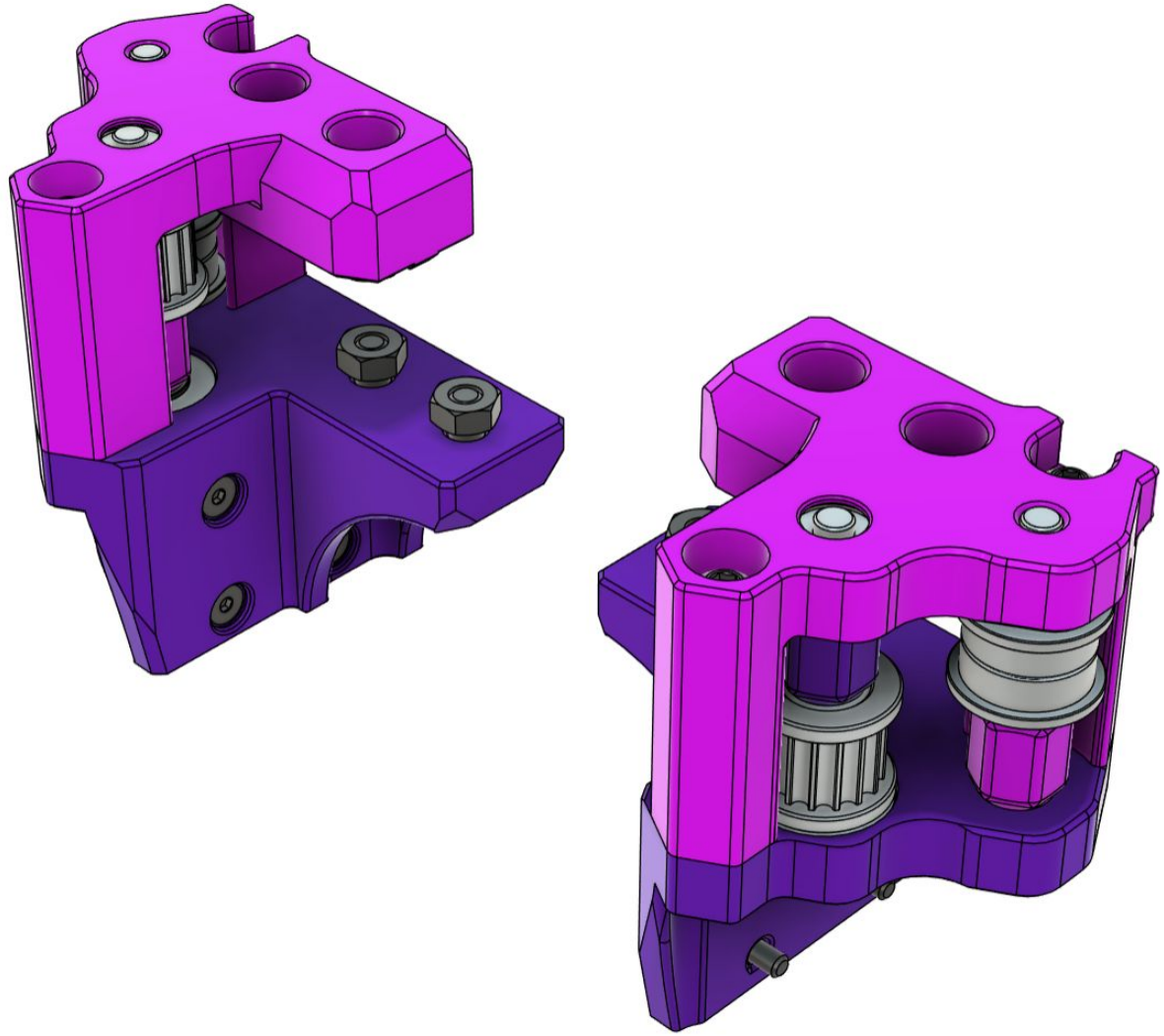


TULIP 2.1 LIVE IDLERS

A Total Pin Mod Gantry with Double Shear & Front Tensioners

VERSION 10-10-2026

“See a PIN and pick it up, and all day long you’ll have a PIN.”

Stanley Howler, Going Postal, by Terry Pratchett

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TULIP MANUAL

This is a Voron V0 gantry Mod that offers Double Shear to the AB motors, Pins to every bearing stack and front tension idlers for easy belt tensioning.
There is also some Printed Tools to help you with installation and would advise you print these off with you parts.

3D PRINTING PROCESS
Fused Deposition Modeling (FDM)

MATERIAL
ABS/ASA

LAYER HEIGHT
Recommended: 0.2mm

EXTRUSION WIDTH
Recommended: Forced 0.4mm

INFILL TYPE
Grid, Gyroid, Honeycomb, Triangle or Cubic

INFILL PERCENTAGE
Recommended: 40%

WALL COUNT
Recommended: 4

SOLID TOP/BOTTOM LAYERS
Recommended: 5

Come Join US - we have cookies
We have a thread on Voron Discord full of friendly and helpful people, so if your stuck and need help or just want to show off your Tulip Build, then come and join us **HERE**

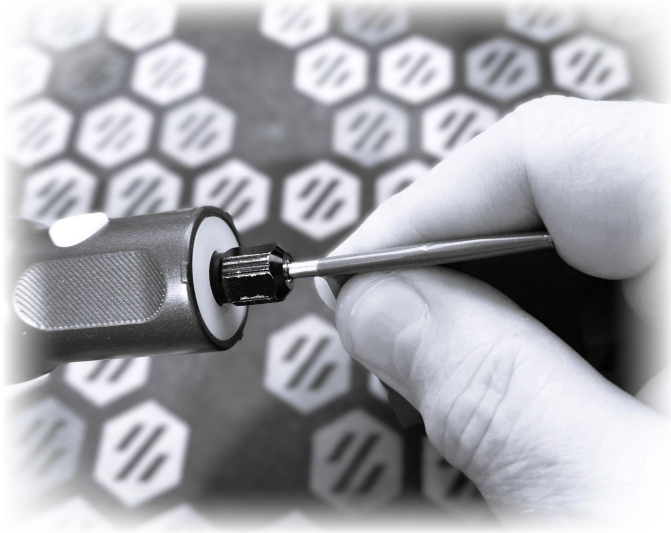
A quick note about your PINS.

Please do a **DRY FIT** to the **IDLERS** and the **BEARINGS.** Some **PINS** may need sanding down to fit nicely

Some **RETAINING COMPOUND** such as **LOCTITE 638** or **641** or similar specs will be required to hold the **SOLID IDLERS** on the **PINS.**

0.5mm SHIMS should be measured prior to assembly. This is to make sure you bearing stacks dont end up to big or too small.
DISCARD any that dont measure **0.47-0.50mm**

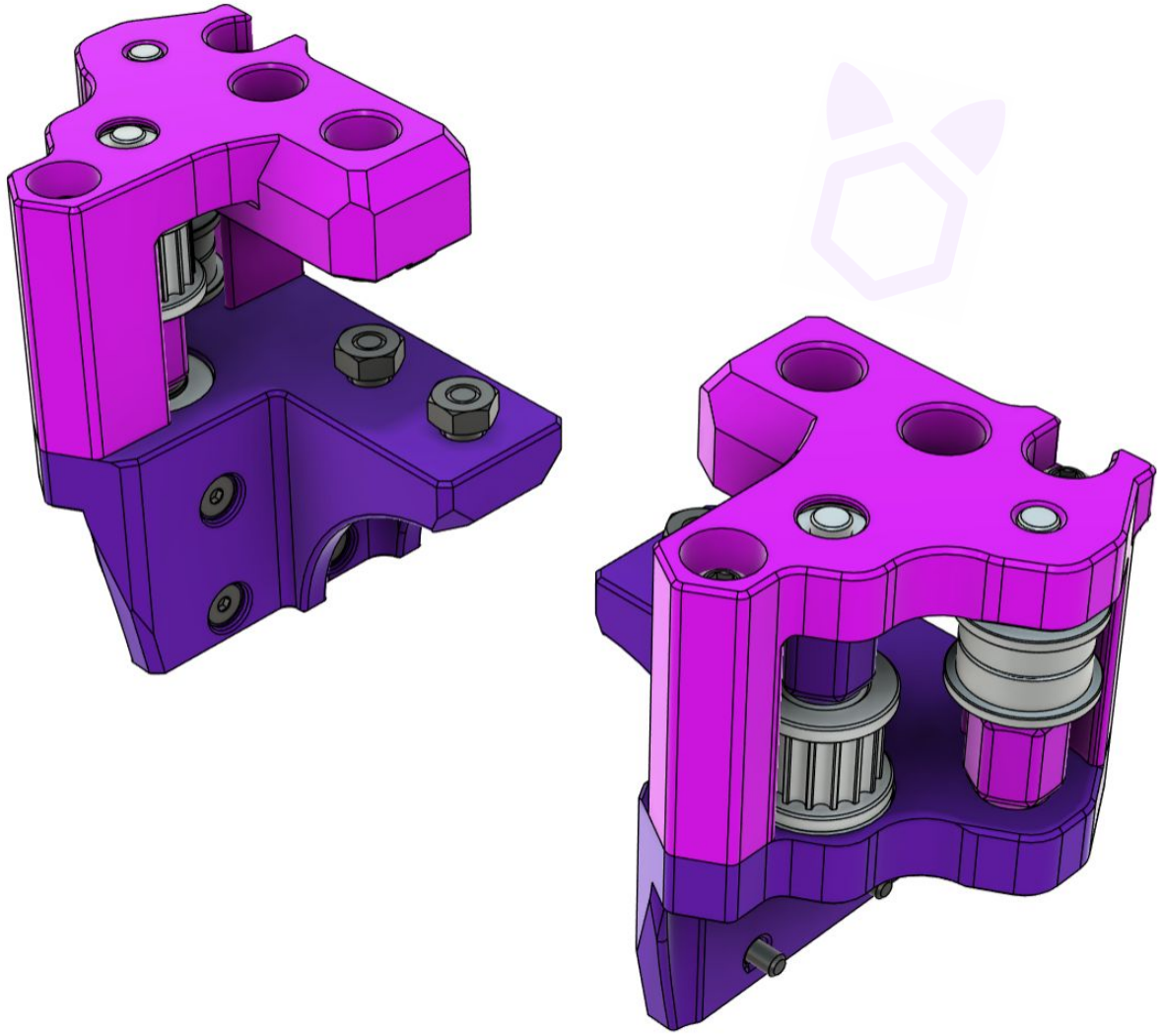
You can use a **DRILL** to put your **PINS** in to sand them down a little bit with some **EMERY CLOTH** or **SANDPAPER.**



Go slow and check often for fitment.
You don't want to go too far.

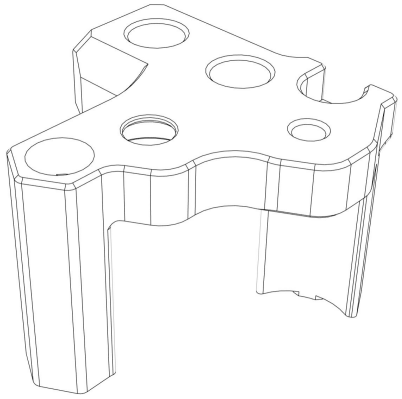
Tulip - Live XY

In this manual we will build our Live XY Joint assemblies.
But first we need to do some prep work.

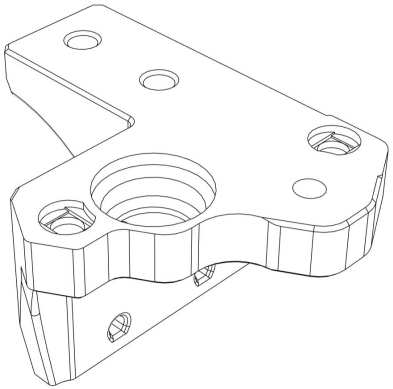


Tulip - Live XY BOM

BOM for building your Special XY joints



XY Joint Top



XY Joint Bottom



Spacers



Screws
M2x6 FHCS x 8
M3x8 BHCS x 8
M3x25 BHCS x 4



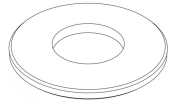
M3 Nuts x 8



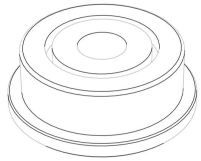
M3 Heat Set Inserts x 4



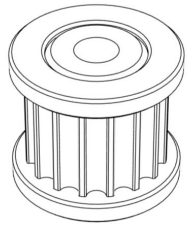
M3x30mm Pins x 4



M3 Shim x 12



F623 Bearing x 8

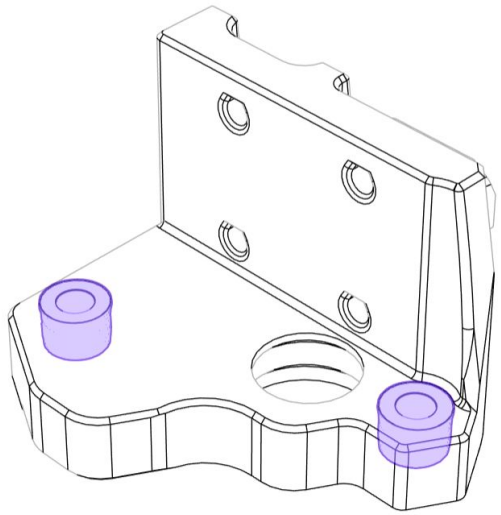


16 Tooth Idler

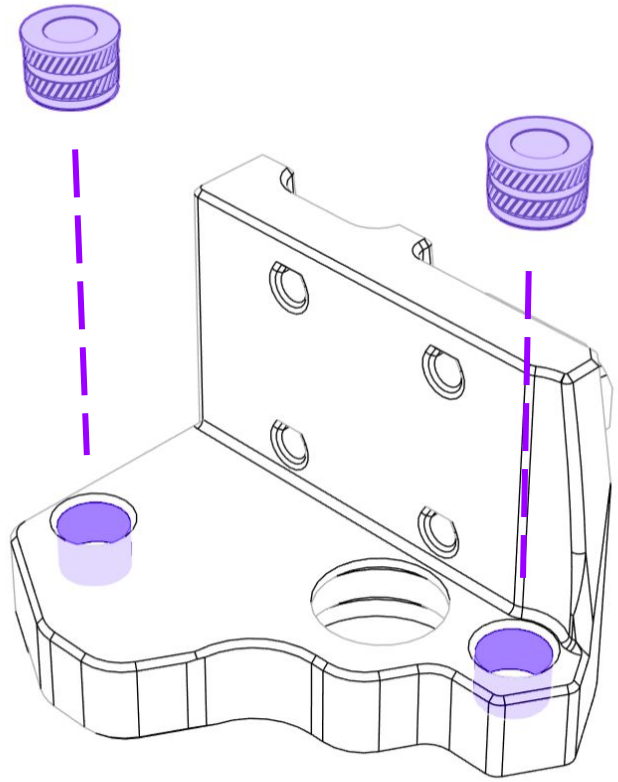
OOO OUCH HOT HOT HOT!

Time to press in some **Voron Spec M3 HEAT SET INSERTS** in to the Bottom XY Joints.
These should be flush when fully seated.

This will need to be done to **BOTH BOTTOM XY JOINTS.**

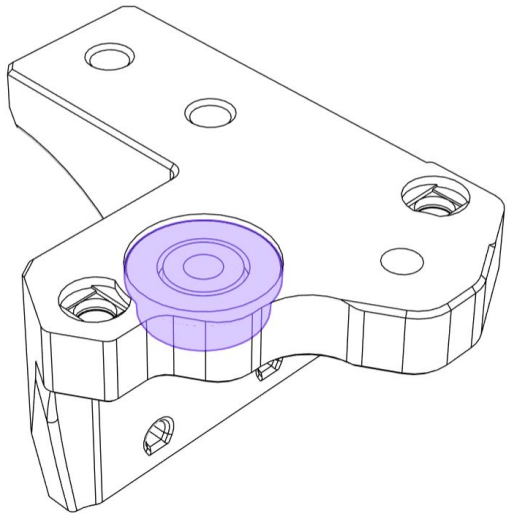


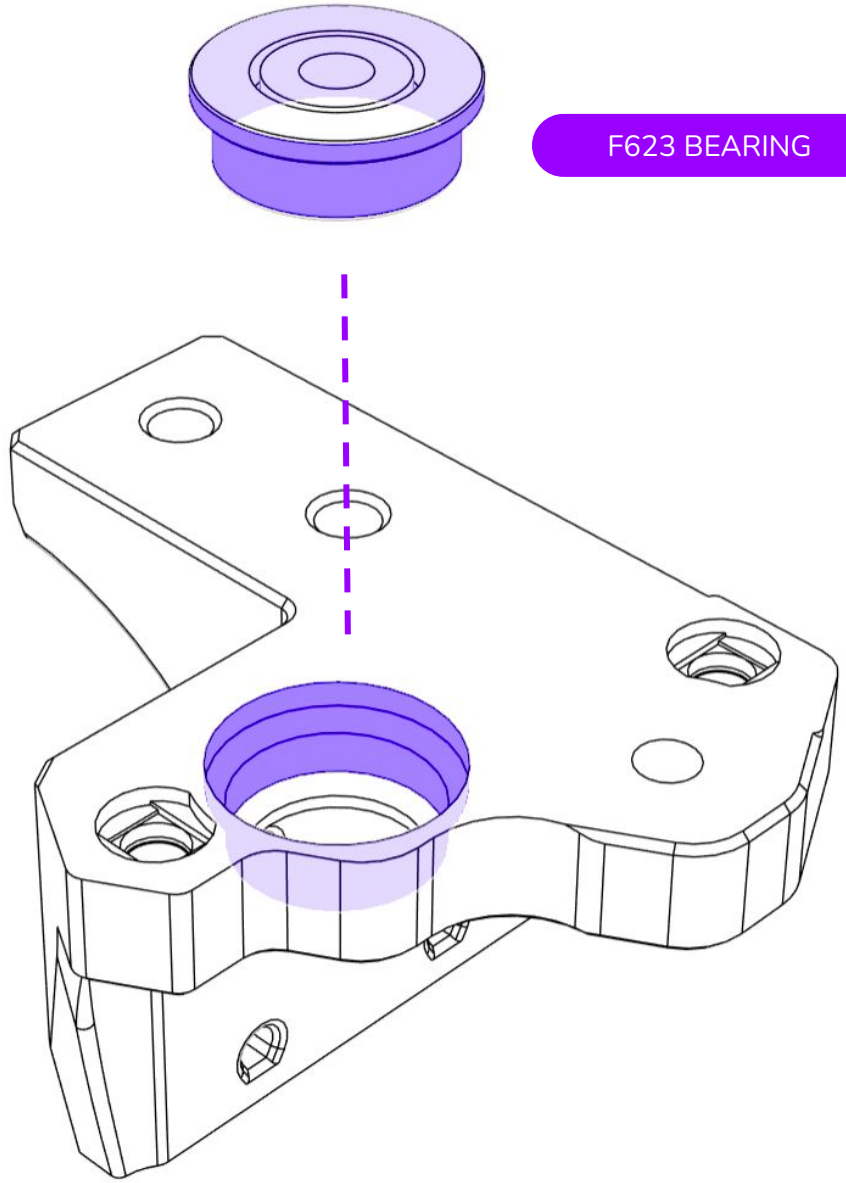
M3 HEAT SET INSERT



YOU SPIN ME RIGHT ROUND BABY

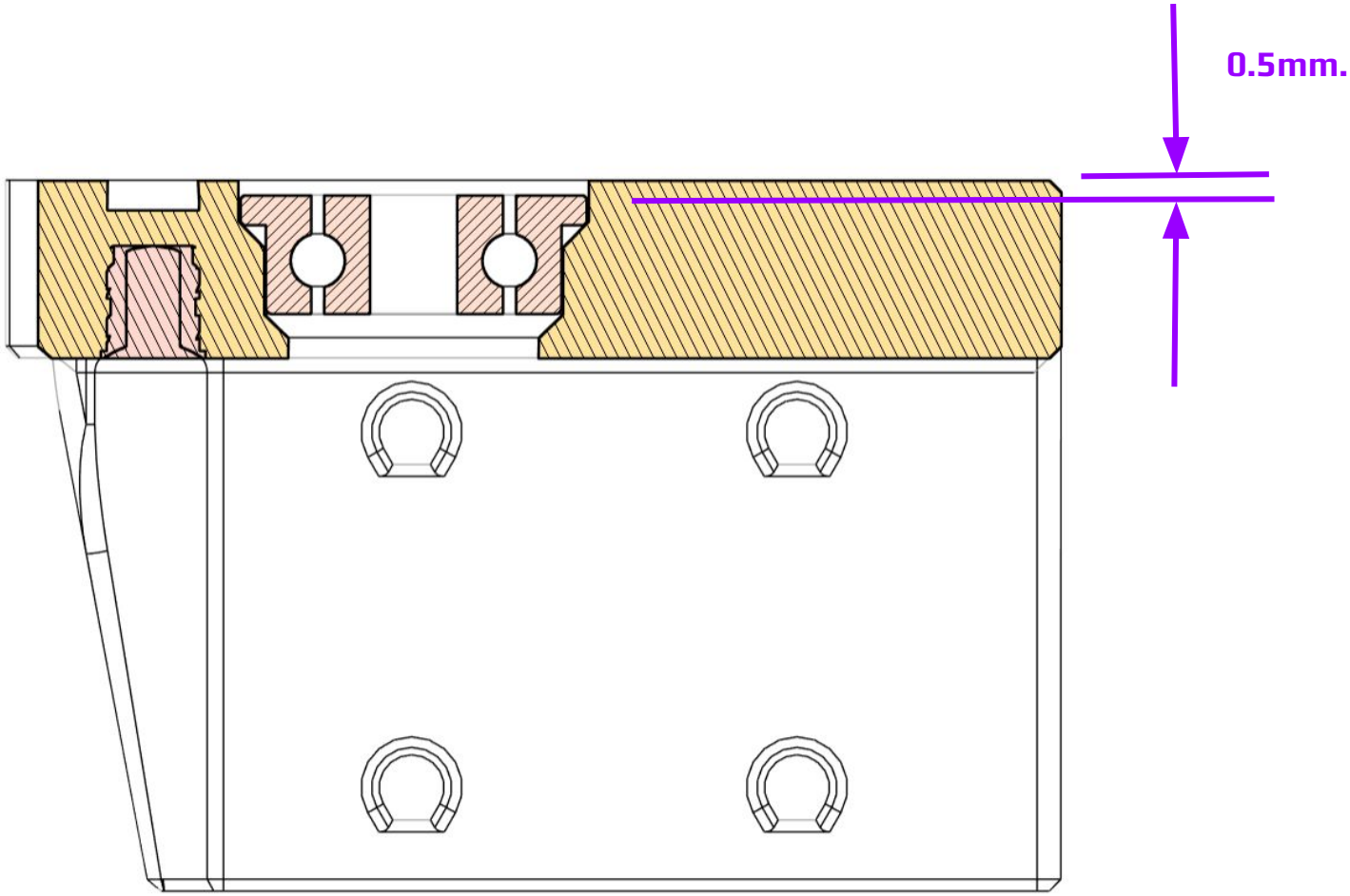
Taking your **F623 BEARINGS** press them into **BOTH** your **BOTTOM XY JOINTS.**





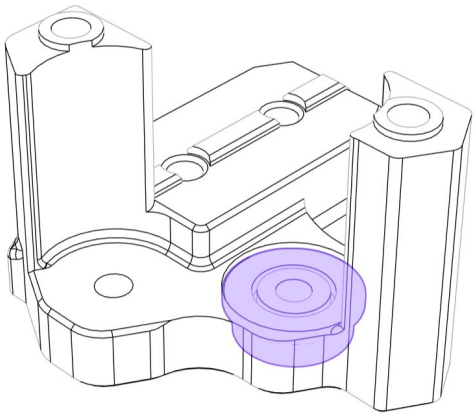
Tulip - Live Bearings

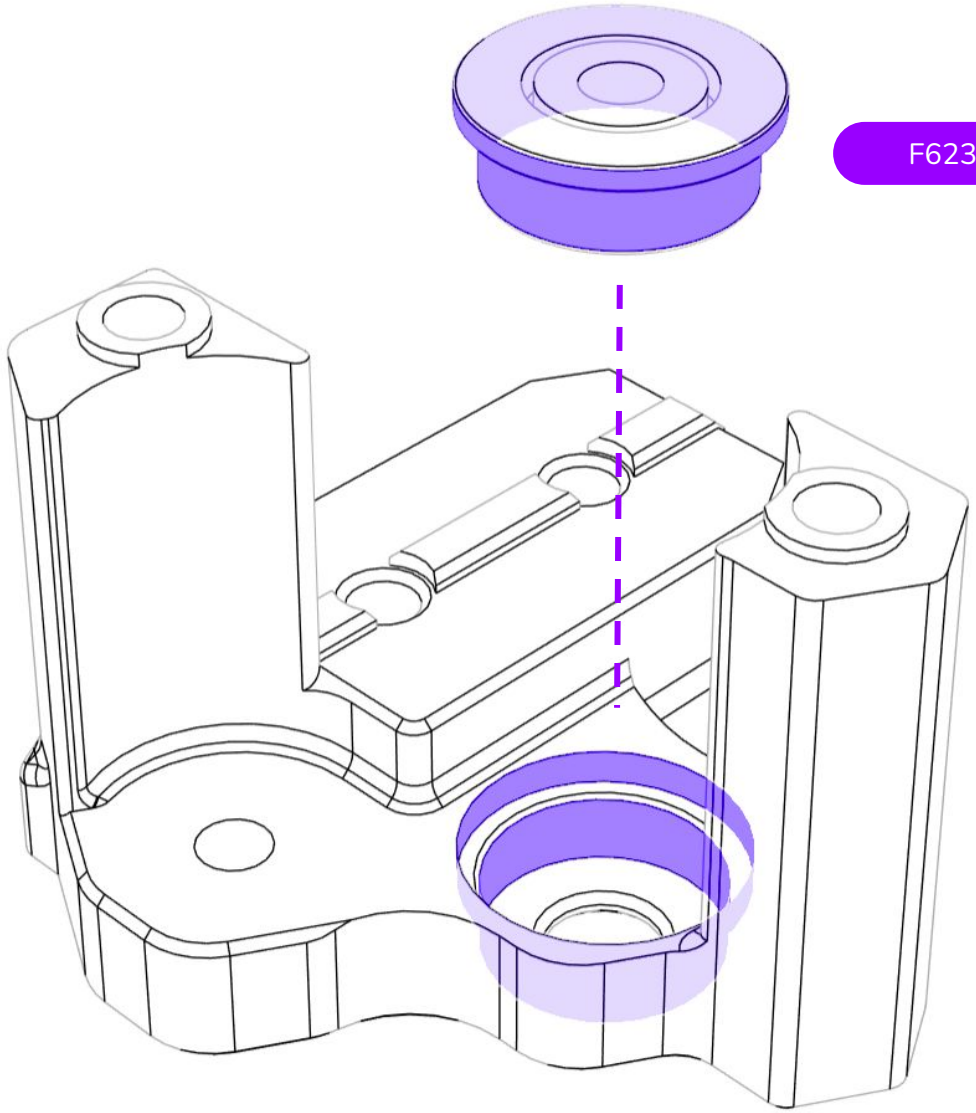
When fully seated the **BEARINGS** should sit just below the **SURFACE** of the **PRINTED PART** by **0.5mm.**



Tulip - Live Bearings

Taking your **F623 BEARINGS** press them into **BOTH** your **TOP XY JOINTS.**

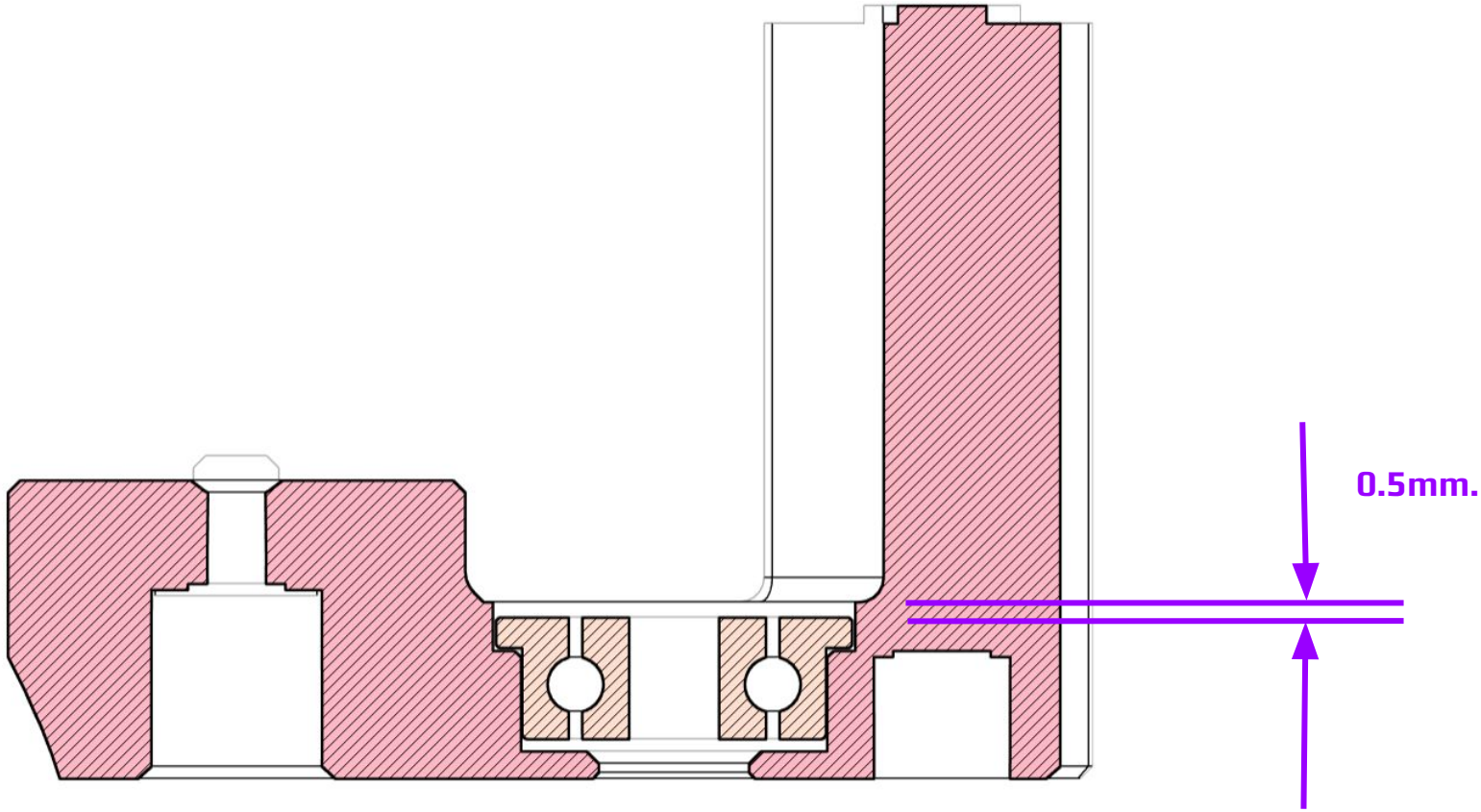




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Tulip - Live Bearings

When fully seated the **BEARINGS** should sit just below the **SURFACE** of the **PRINTED PART** by **0.5mm.**

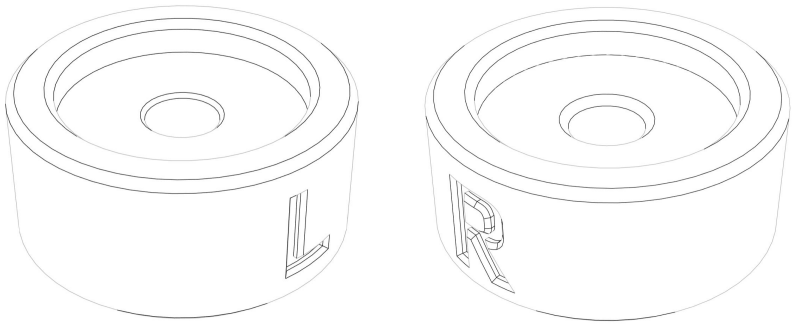


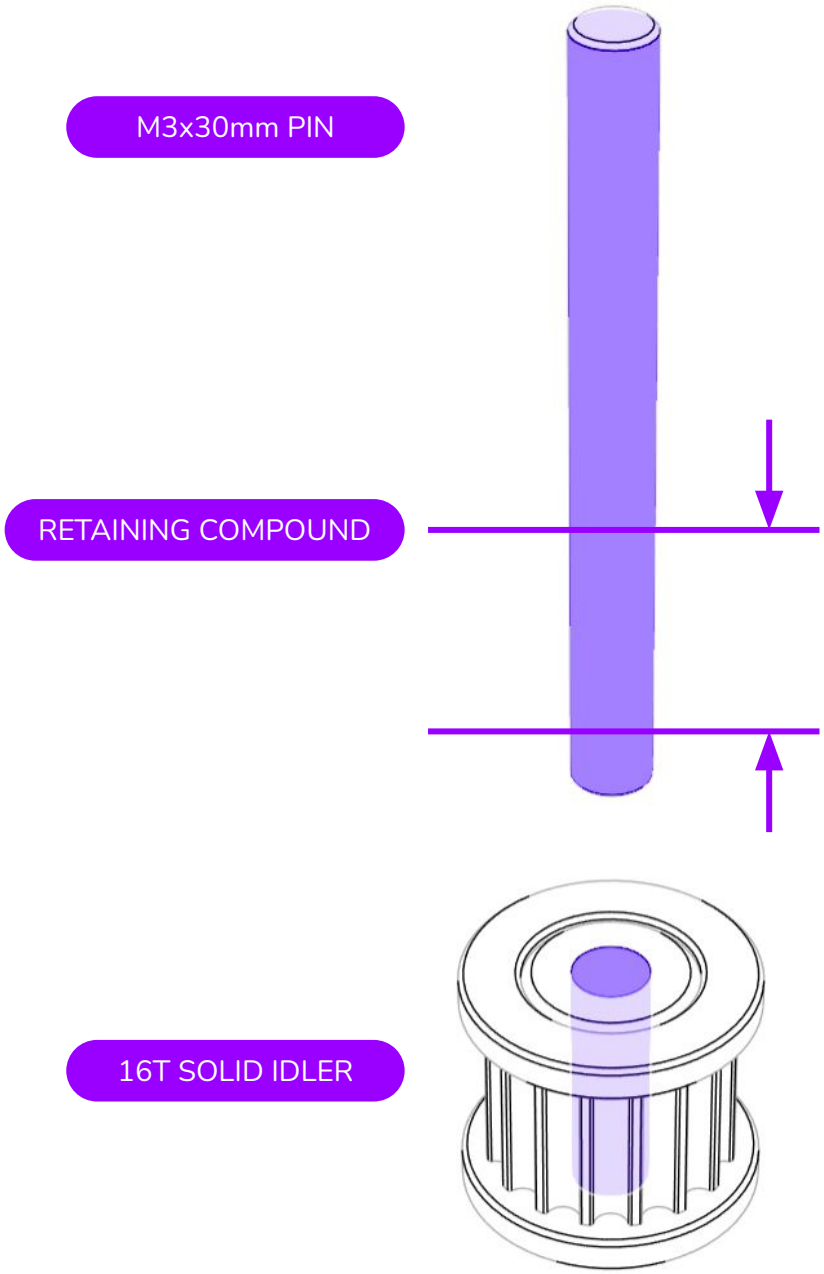
Tulip - Live Solid Idler

STICK WITH ME HERE

Included in the **STL's** are some **TOOLS** to help you with **building TULIP MOD.**
One such tool is the **LIVE JIGS.**
You'll find a nice little mark for **L** and **R** so it's easier to keep track.
Apply a little **RETAINING COMPOUND** to where the idler will roughly sit on the **PIN,** Slide on the **SOLID IDLER** and give it a little **TWIST** (This will help distribute the **RETAINING COMPOUND**) and pop the assembly into their corresponding **LIVE JIG.**

Its is advised to allow these to sit for **24 HOURS** to fully cure.

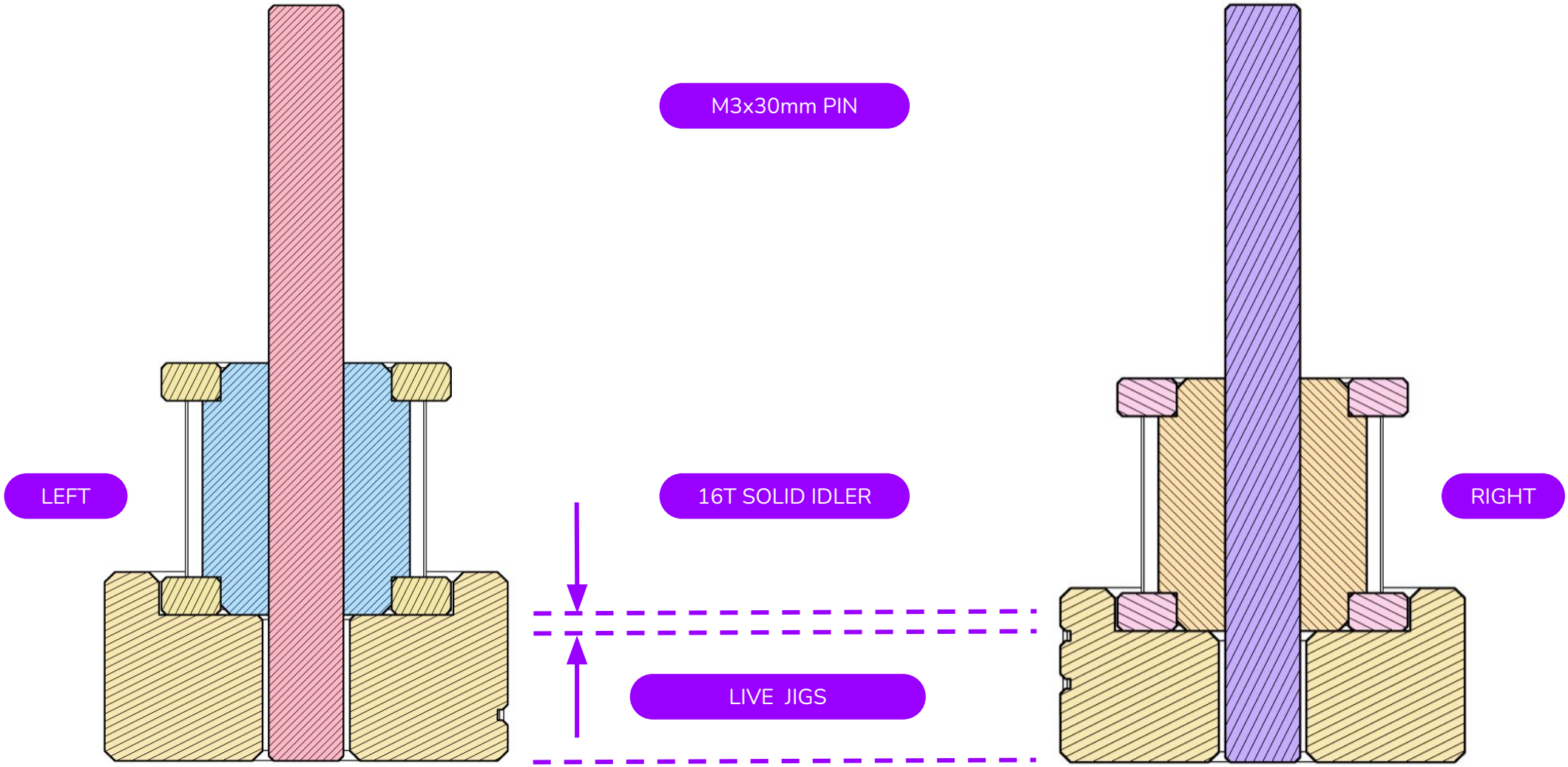




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Tulip - Live Solid Idler

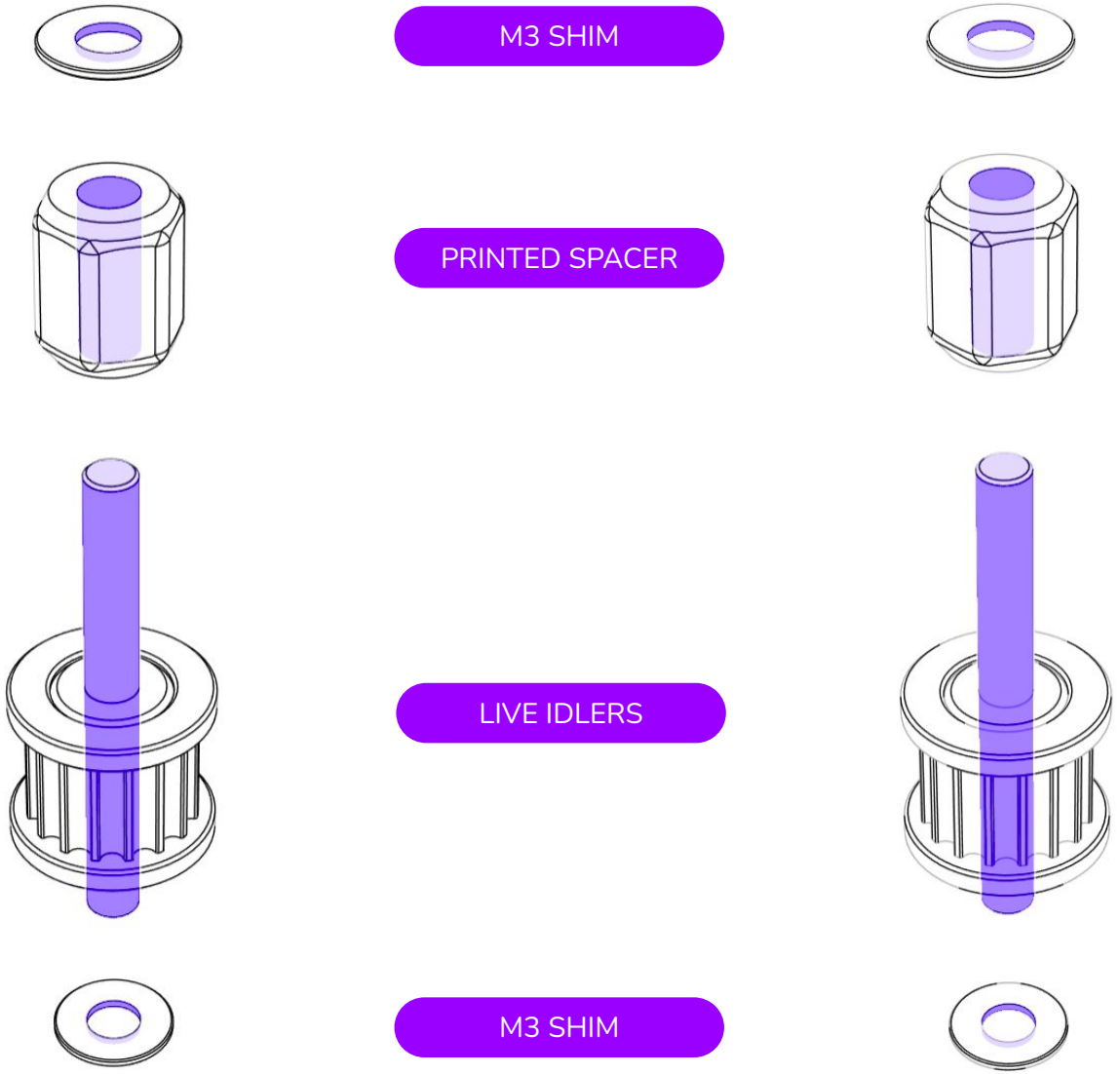
Make sure to sit these on a flat surface for **24 HOURS** to fully cure.



14

TIME TO STACK IT UP BABY

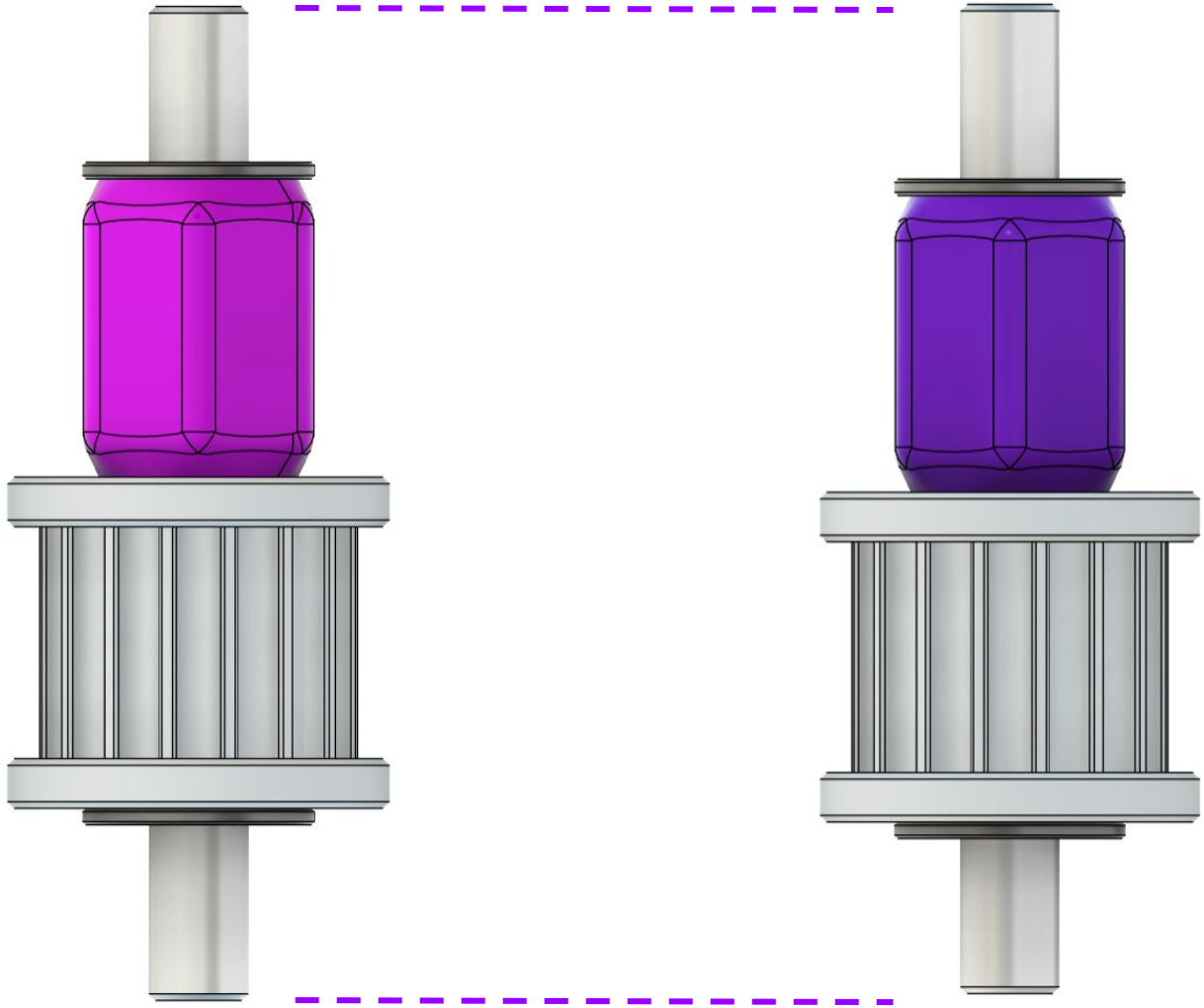
Adding your **PRINTED SPACERS** followed by **M3 SHIMS** to the stacks to complete.



Tulip - Live Solid Idlers

They should now look like this

LEFT IDLER



RIGHT IDLER

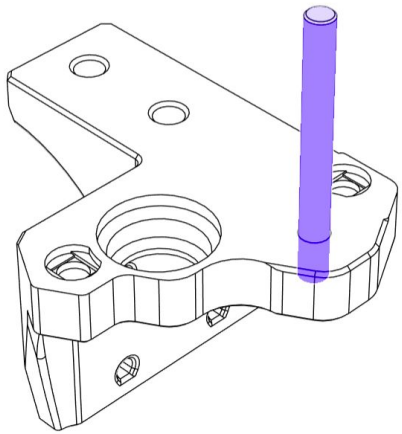
16

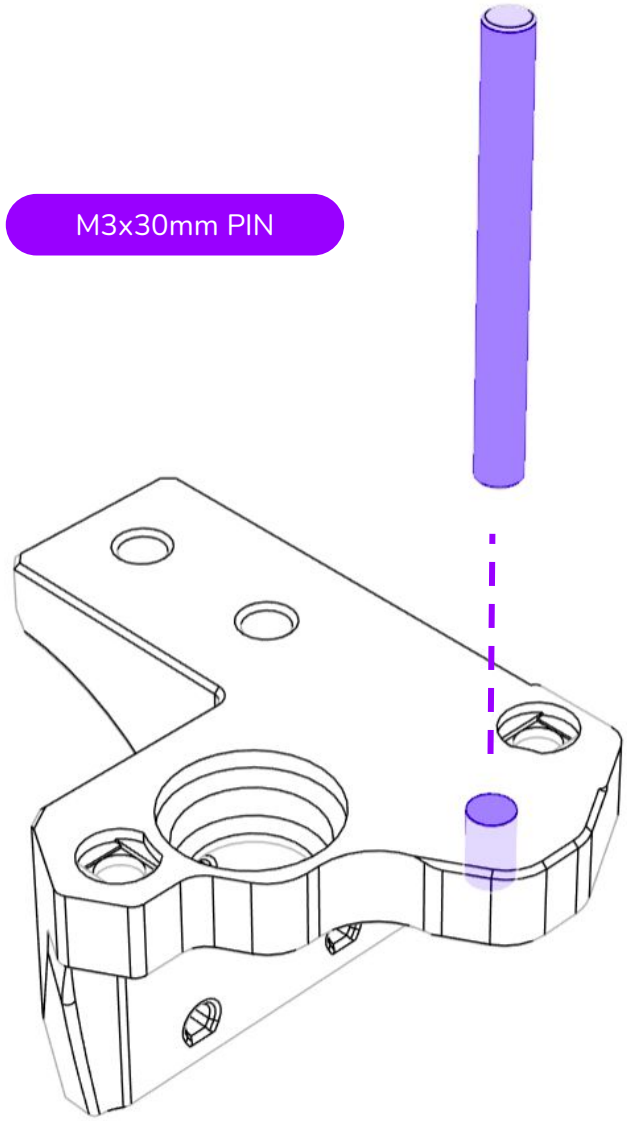
Tulip - Right XY Joint

JUST STICK A PIN IN IT!

Take the **RIGHT LOWER XY** Joint and an **M3x30mm PIN** and press into the corresponding hole.

It will be quite tight and this is desired as we dont want these moving.





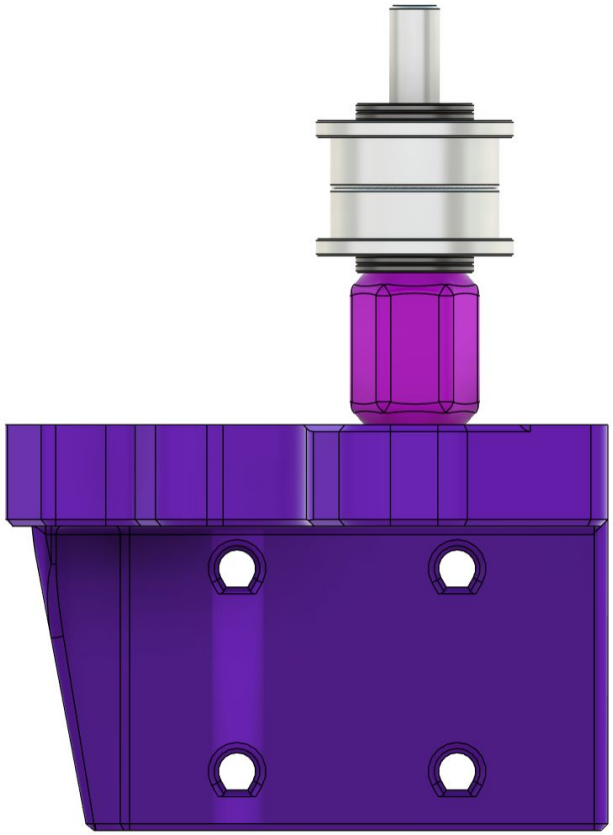
17

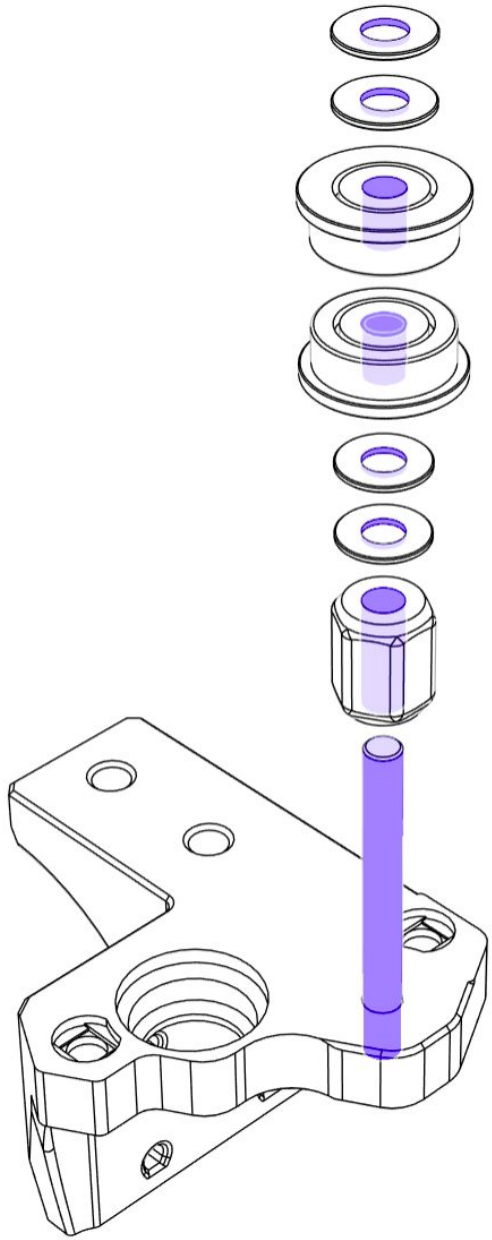
Tulip - Right XY Joint

Time to Stack it up!

Lets add our bearing stack to the **M3x30mm PIN.**

IT SHOULD LOOK LIKE THIS WHEN FINISHED.

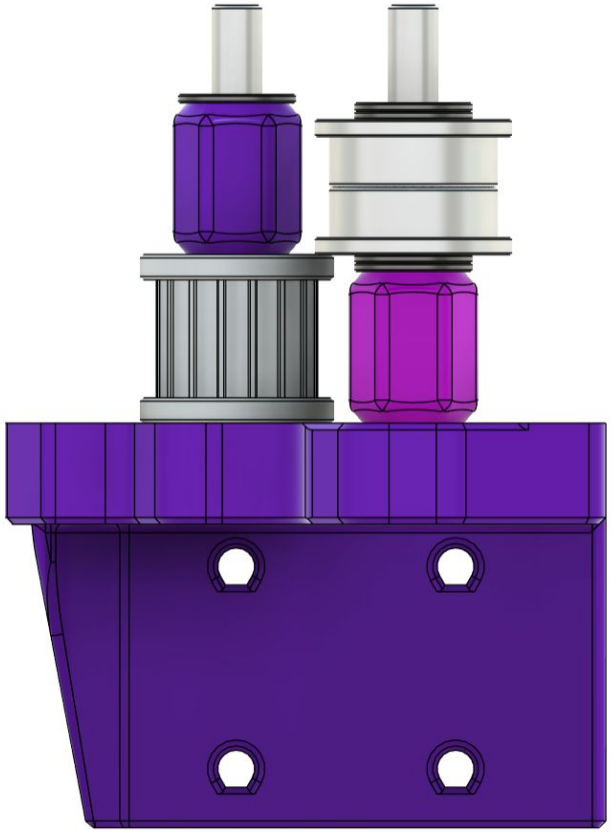




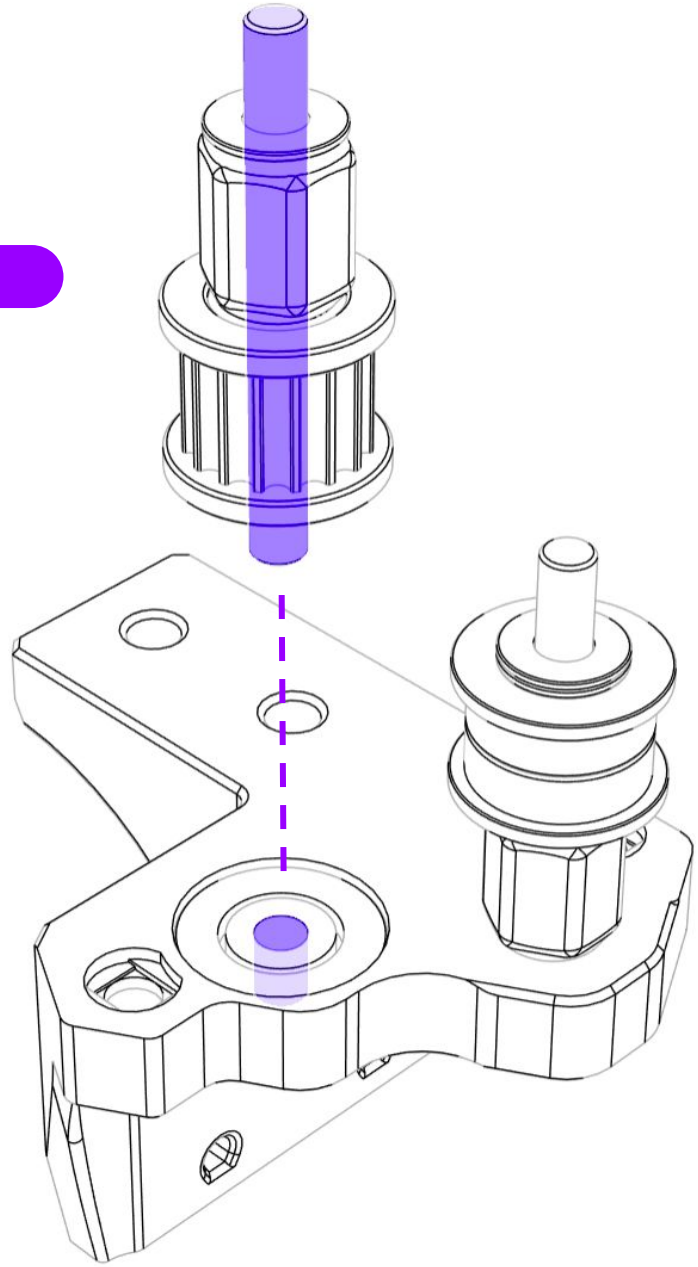
18

GOING LIVE IN 5... 4... 3... 2... 1!

When slotting the **LIVE IDLERS** into the bearing make sure you dont loose your **SHIM** off the bottom.

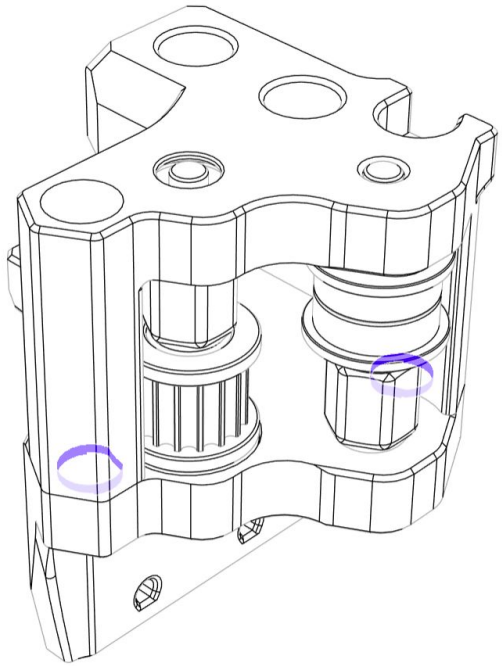


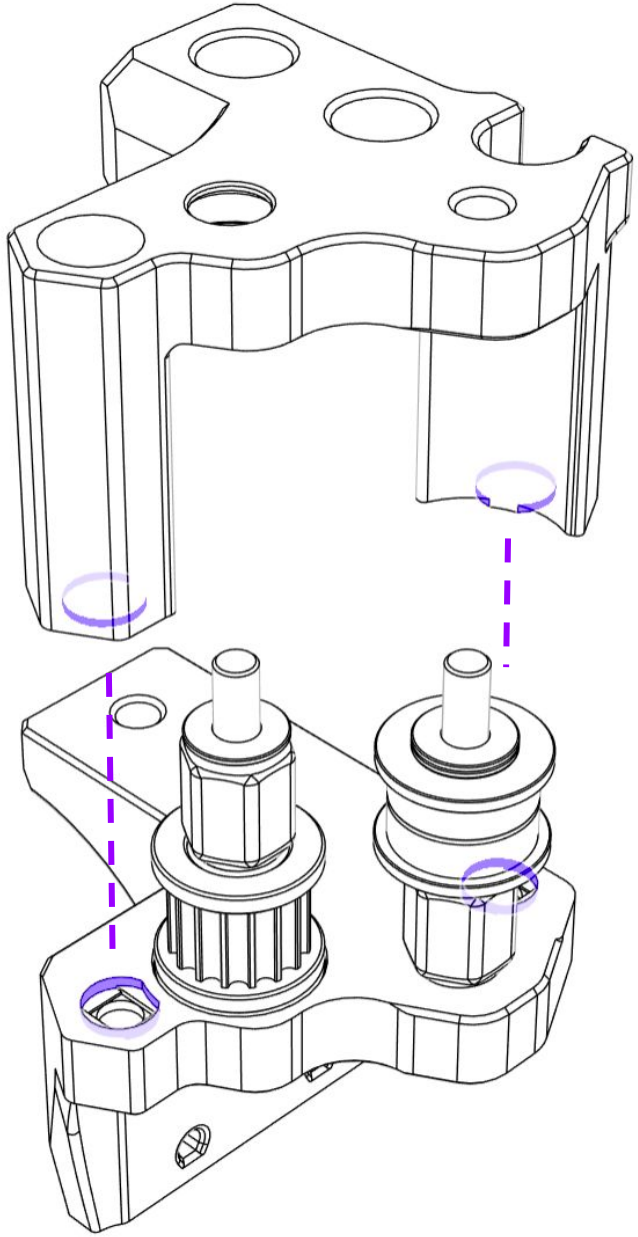
LIVE IDLER



Tulip - Right XY Joint

Time to fit the **TOP XY JOINT.**
You may have noticed there are two keys to aid location.

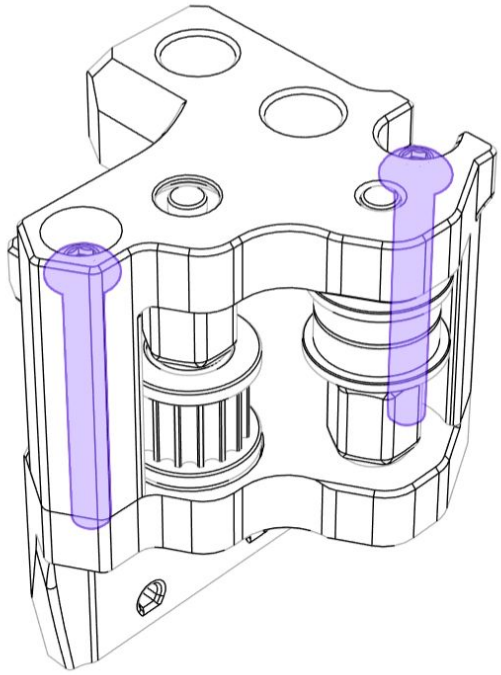




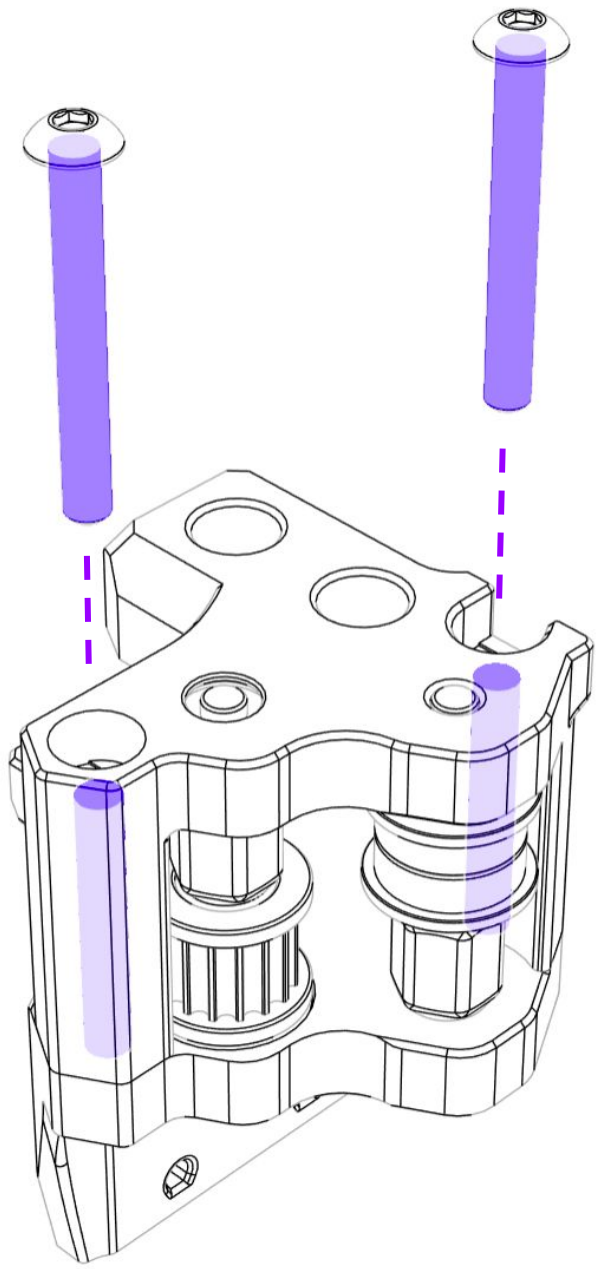
20

Tulip - Right XY Joint

Take two **M3x25 BHCS** screws to lock the **TOP** and **BOTTOM XY JOINTS** together.



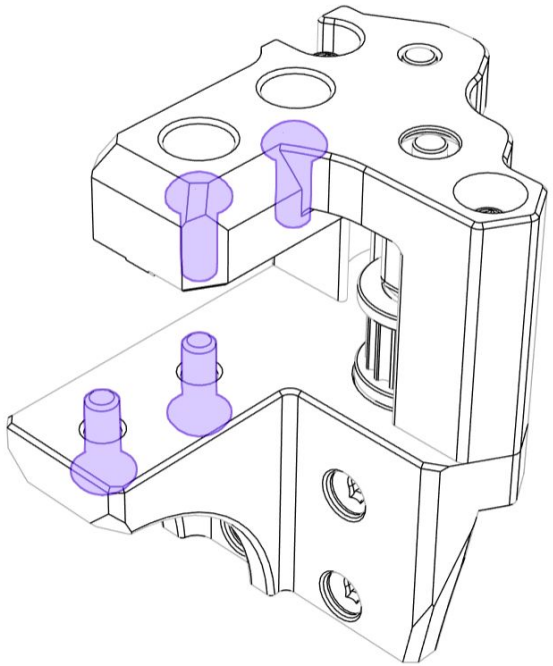
M3X25 BHCS



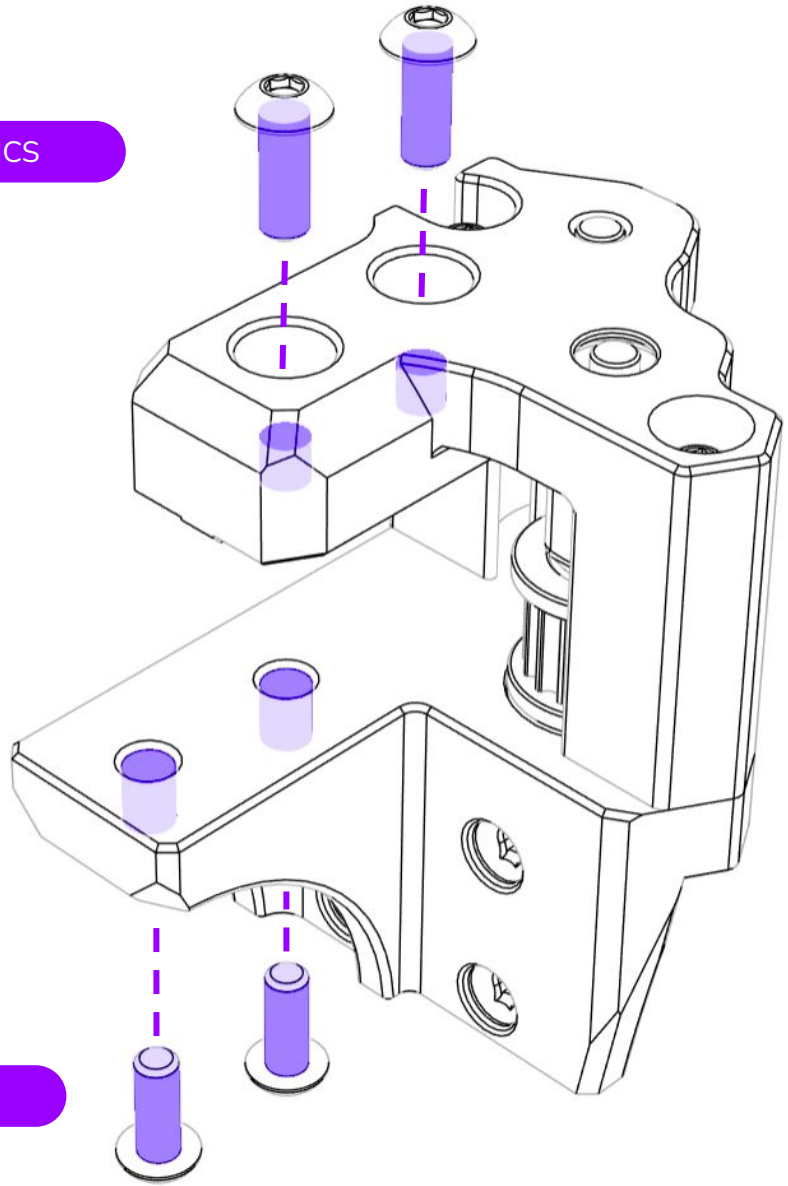
21

Tulip - Right XY Joint

Take four **M3x8 BHCS** screws and add to the **TOP** and **BOTTOM XY JOINTS**.



M3X8 BHCS

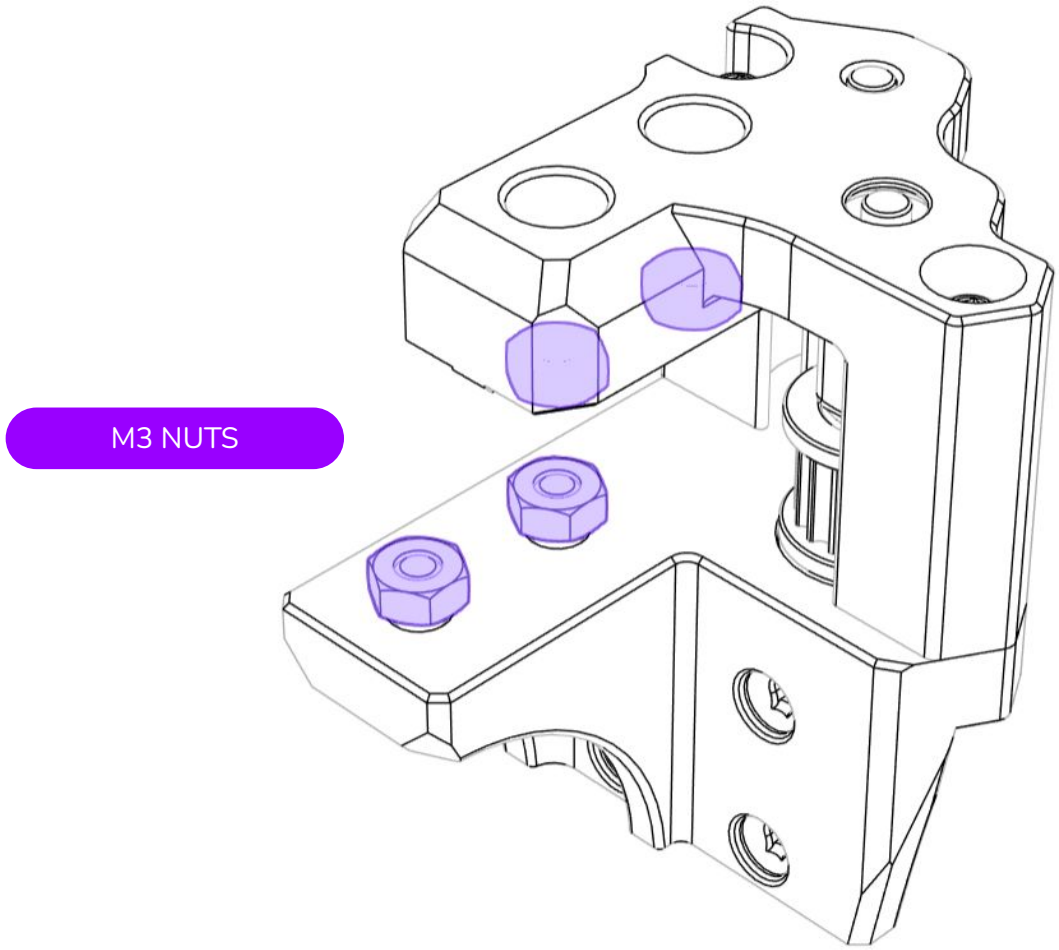


M3X8 BHCS

22

Tulip - Right XY Joint

Take four **M3 NUTS** and add to the **TOP** and **BOTTOM XY JOINTS**.
You are now ready to install this onto your gantry.



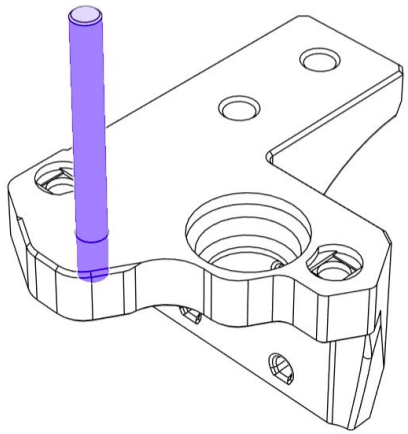
23

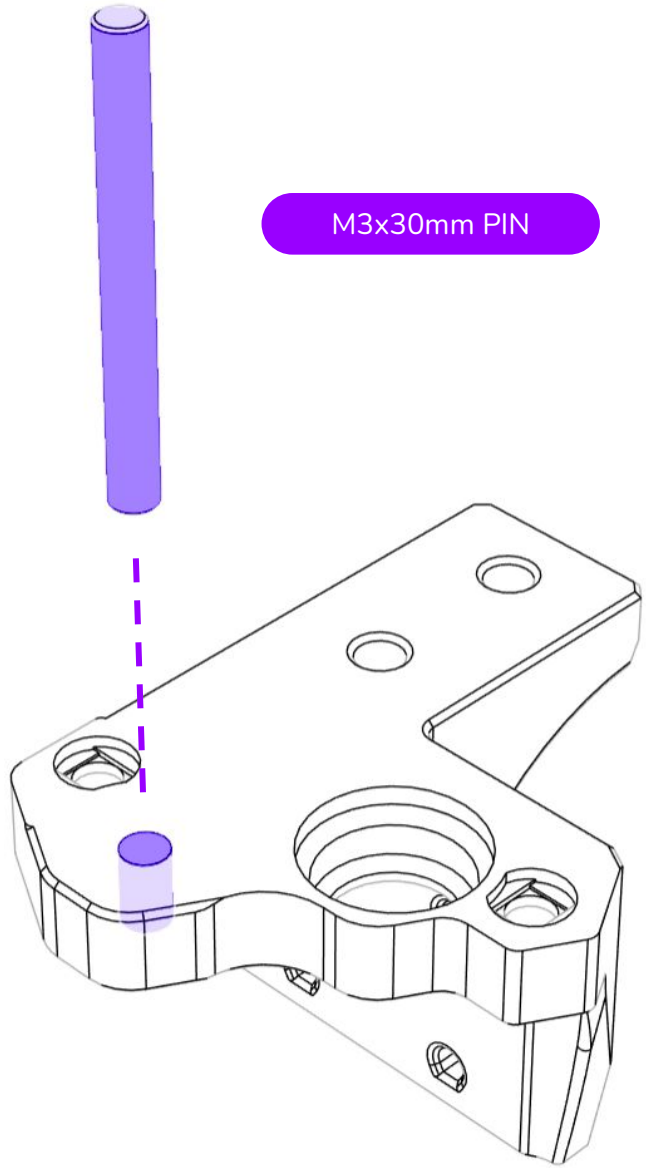
Tulip - Left XY Joint

JUST STICK A PIN IN IT!

Take the **LEFT LOWER XY** Joint and an **M3x30mm PIN** and press into the corresponding hole.

It will be quite tight and this is desired as we dont want these moving.





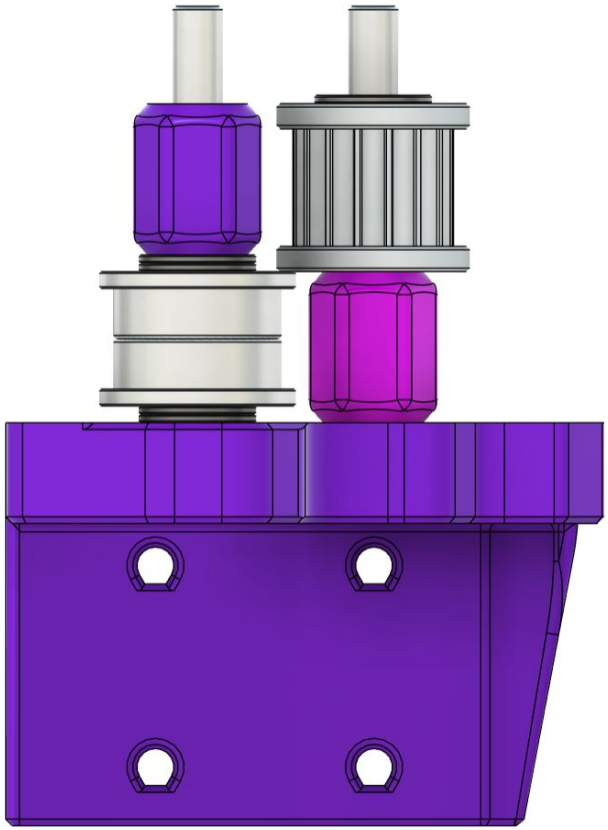
24

Tulip - Left XY Joint

Time to Stack it up!

Lets add our bearing stack to the **M3x30mm PIN.**

IT SHOULD LOOK LIKE THIS WHEN FINISHED.



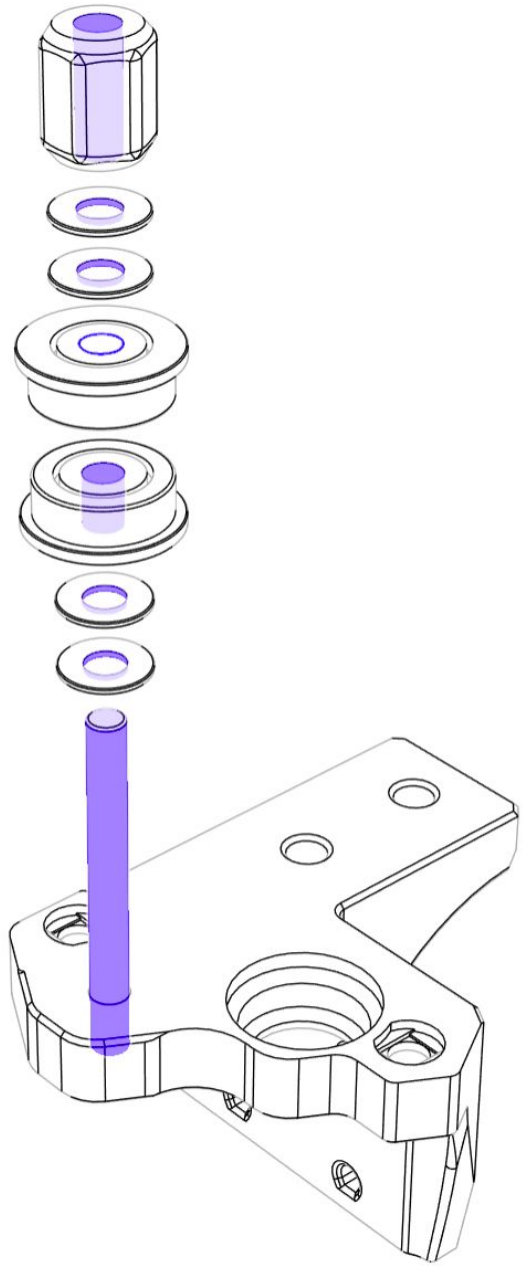
PRINTED SPACER

M3 SHIM

F623 BEARING

M3 SHIM

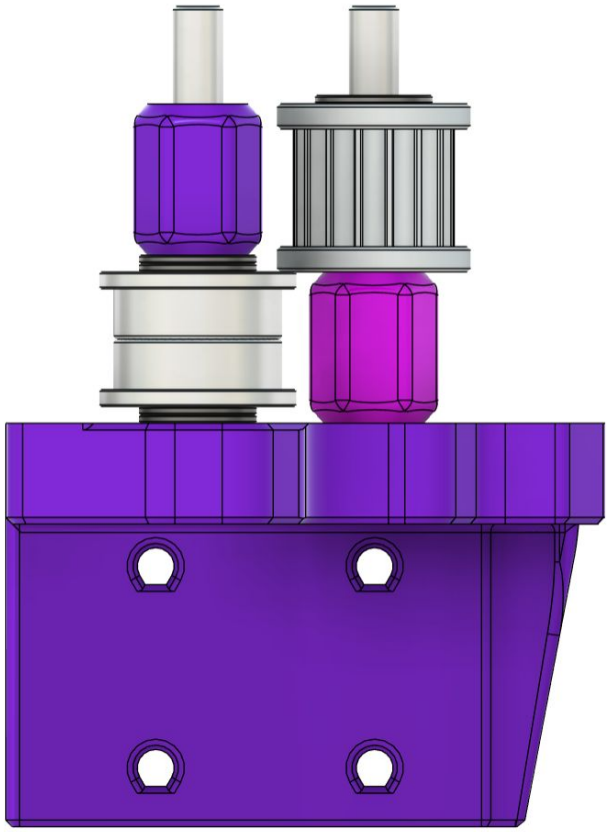
M3x30mm PIN

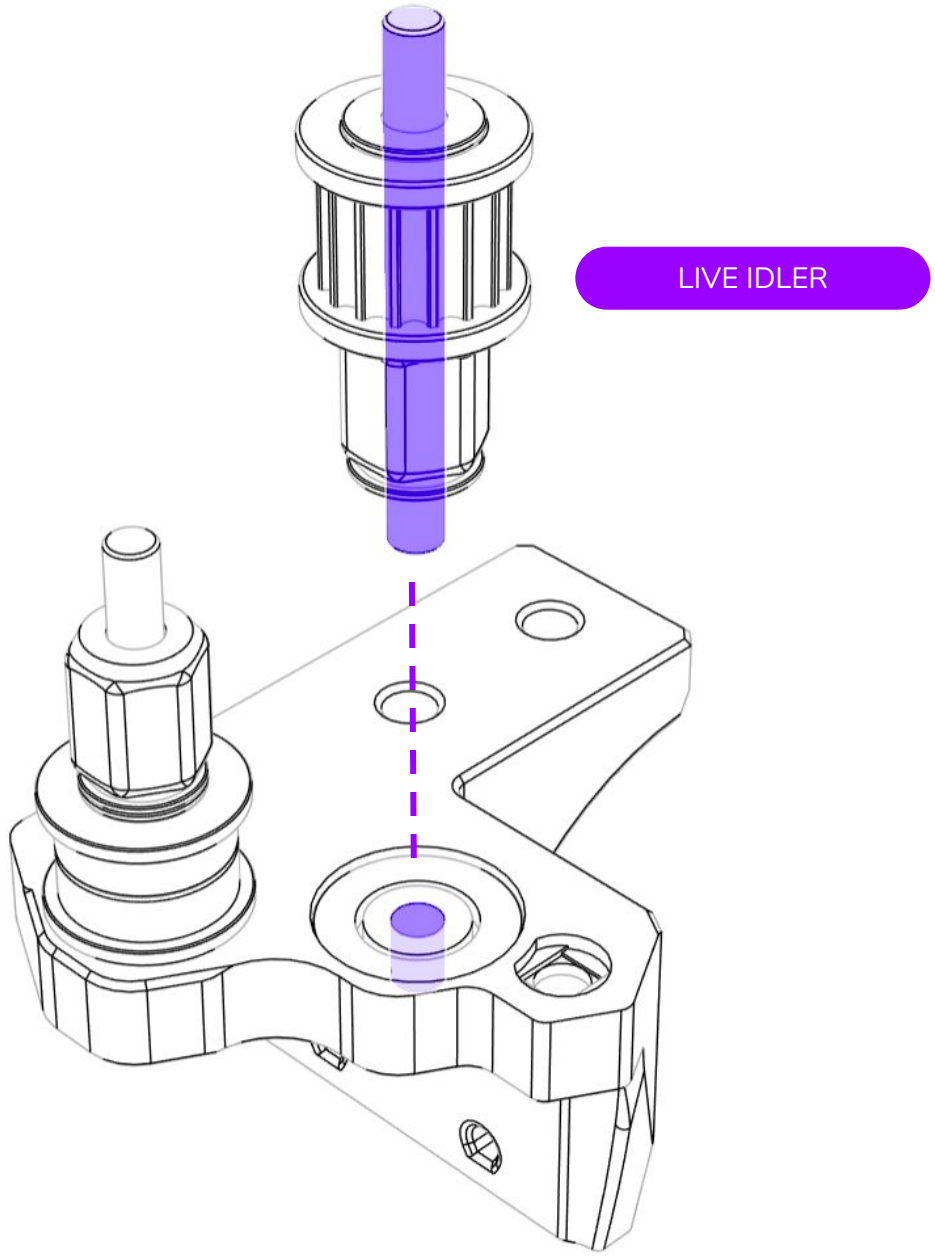


25

GOING LIVE IN 5... 4... 3... 2... 1!

When slotting the **LIVE IDLERS** into the bearing make sure you dont loose your **SHIM** off the bottom.

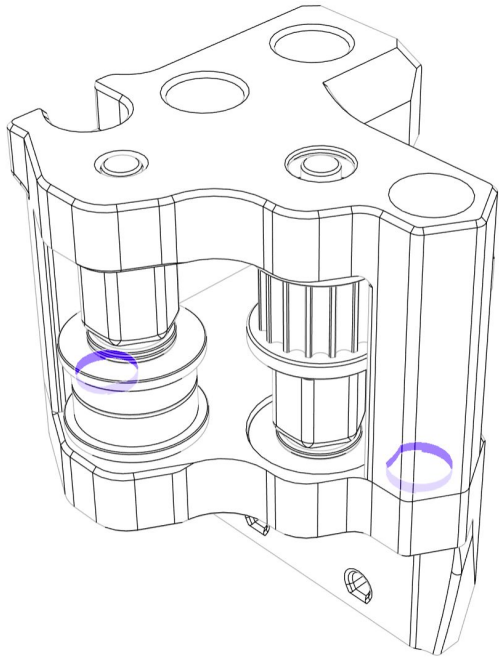


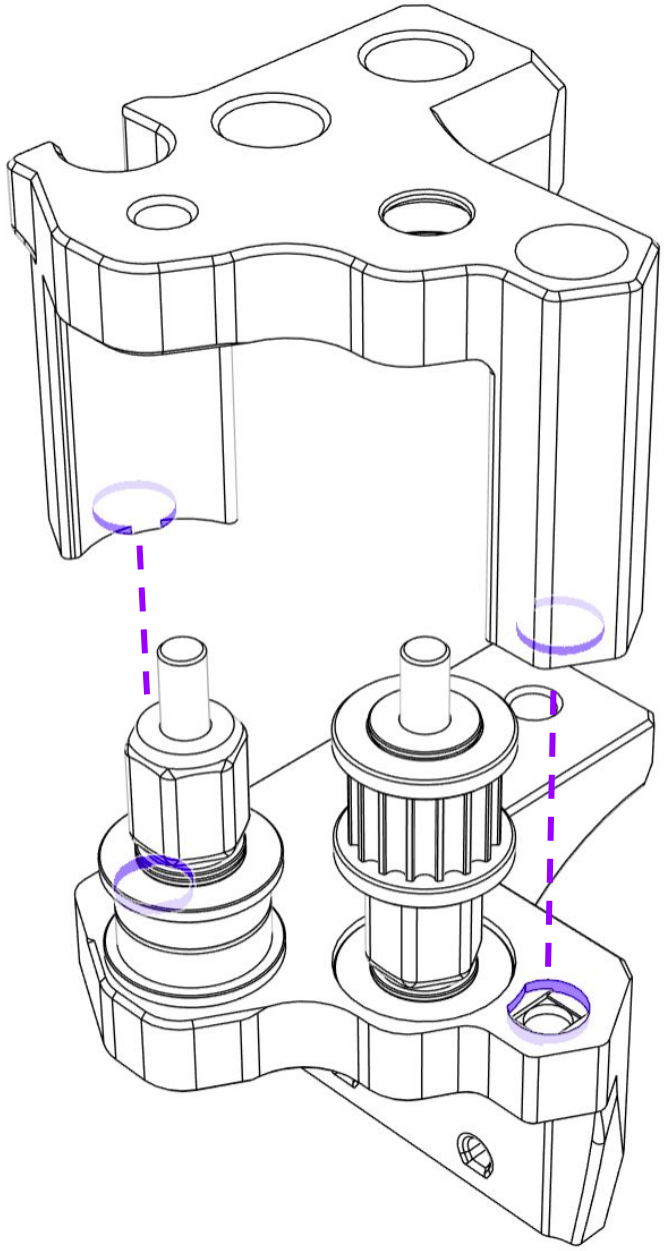


26

Tulip - Left XY Joint

Time to fit the **TOP XY JOINT.**
You may have noticed there are two keys to aid location.

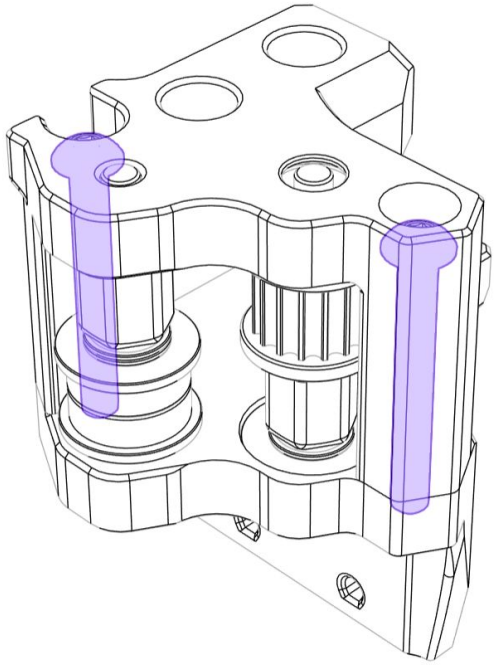


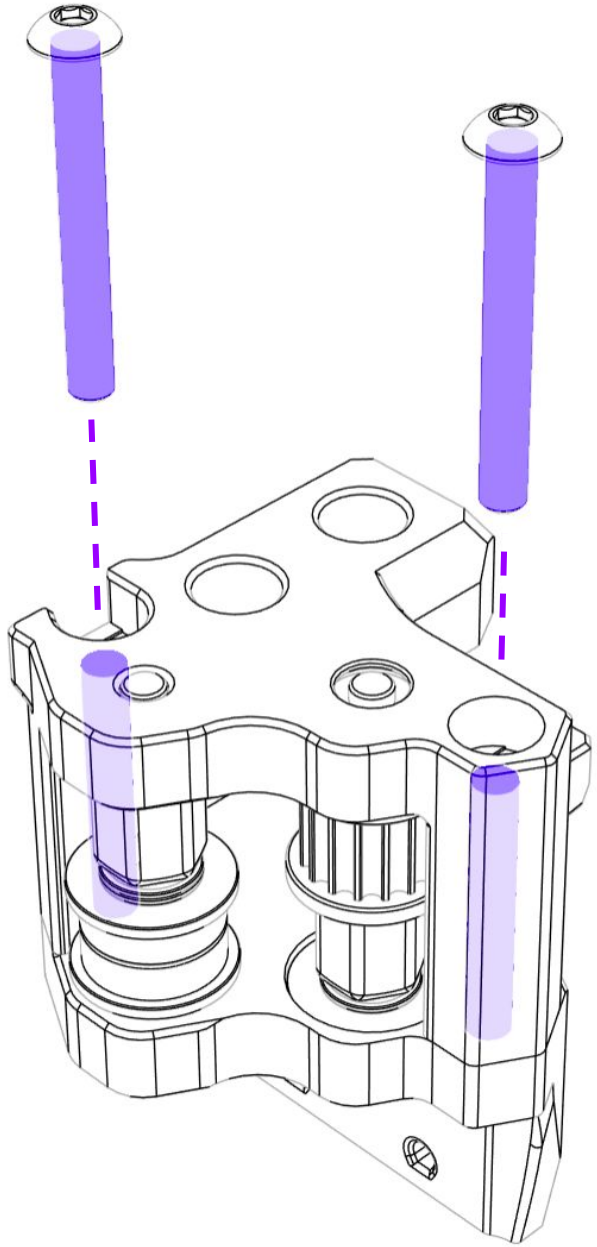


27

Tulip - Left XY Joint

Take two **M3x25 BHCS** screws to lock the **TOP** and **BOTTOM XY JOINTS** together.



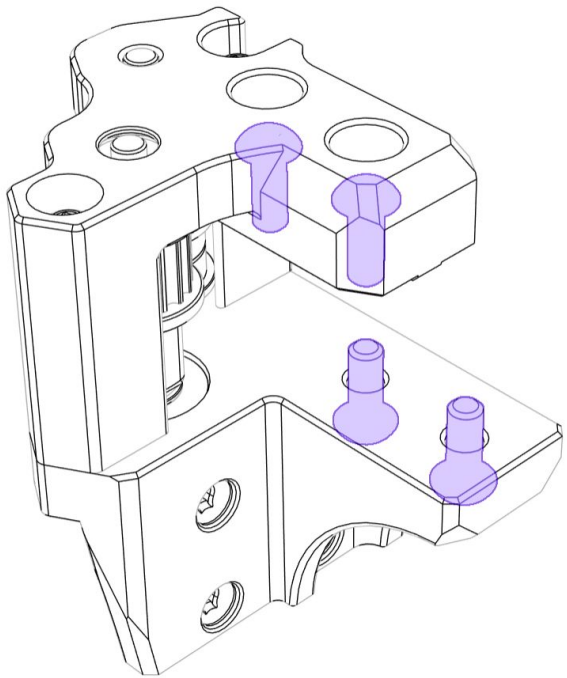


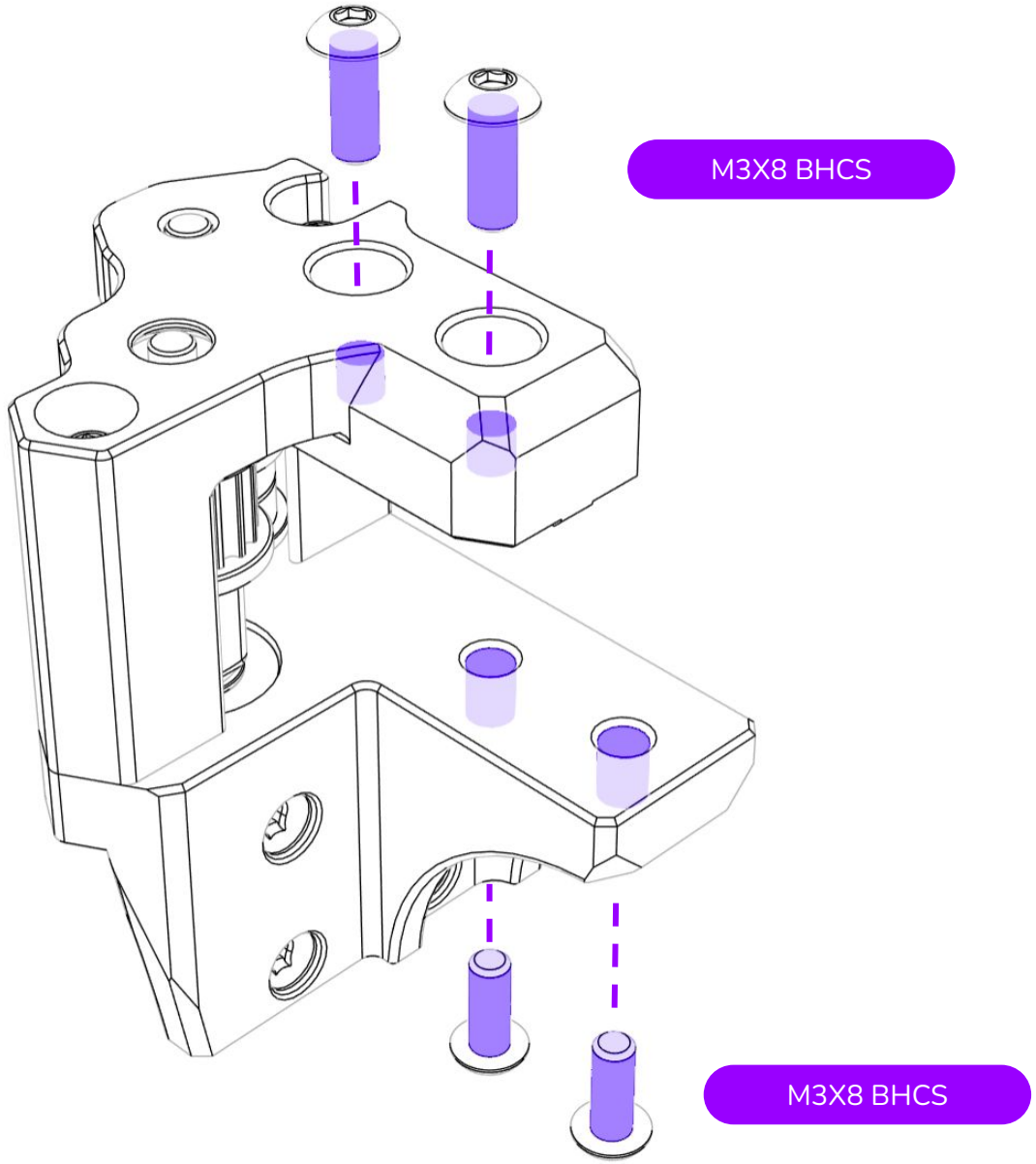
M3X25 BHCS

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Tulip - Left XY Joint

Take four **M3x8 BHCS** screws and add to the **TOP** and **BOTTOM XY JOINTS**.

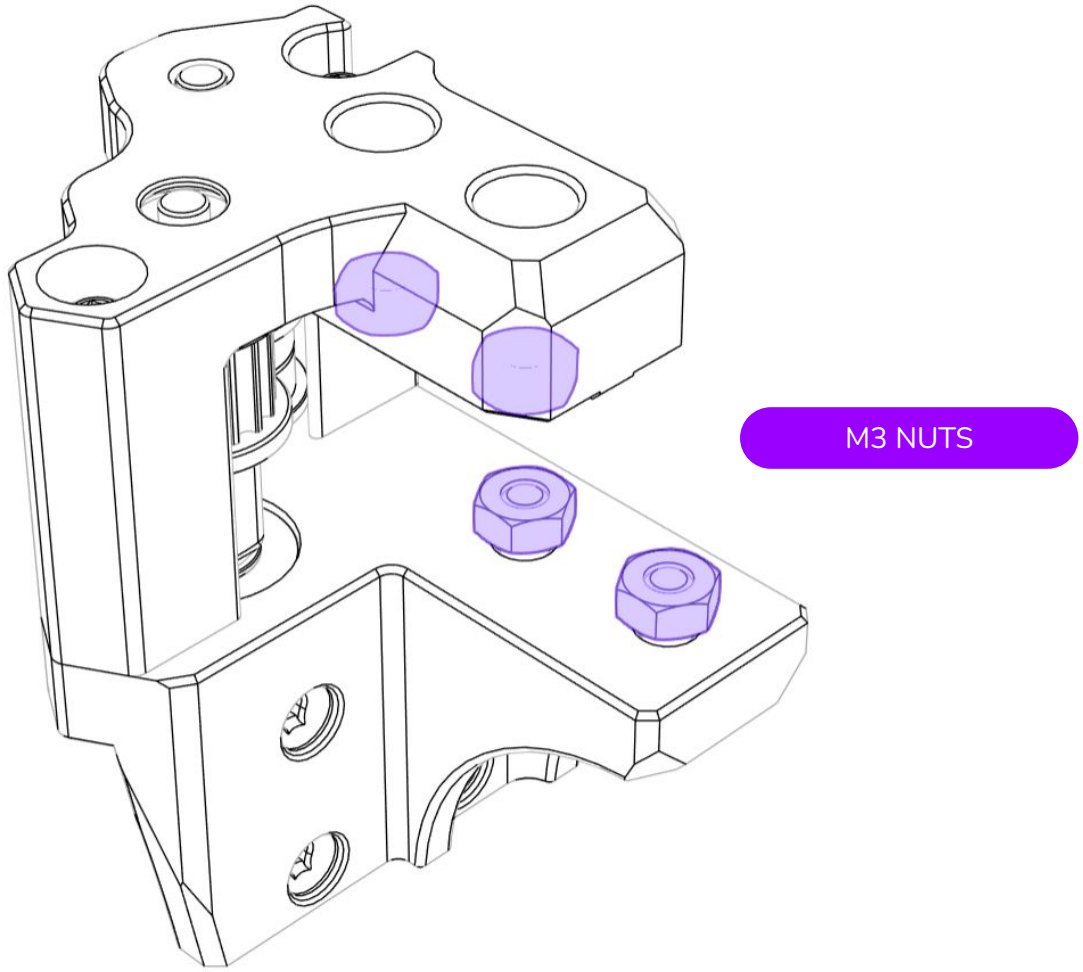




29

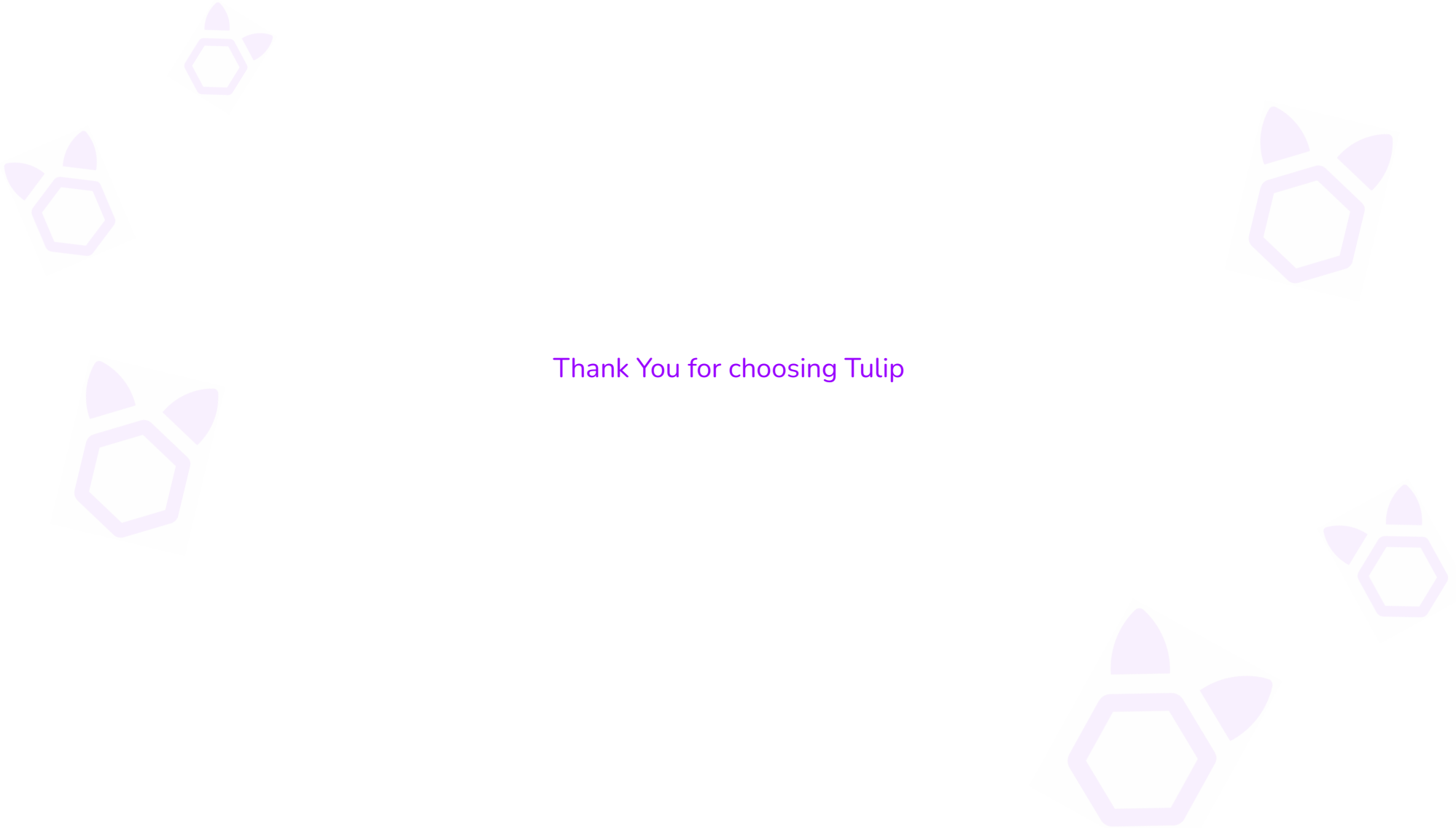
Tulip - Left XY Joint

Take four **M3 NUTS** and add to the **TOP** and **BOTTOM XY JOINTS**.
You are now ready to install this onto your gantry.



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Tulip - End



Thank You for choosing Tulip

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